Marketing Campaign Analysis — Phase 4: CLV & Response Propensity

Omar Touzani · Two models: regression to predict customer value, classifier to predict campaign response, combined into a targeting priority matrix.

I trained two model families. The regression predicts a customer's total spend using only demographic, tenure, recency, web-visit, and historical-campaign features — no purchase counts or per-category spend, which would have leaked the target. The classifier predicts whether a customer accepts the latest campaign, using behavioural features as well. Models were selected by per-class behaviour first, not headline metrics.

CLV regression — predicting customer value

Model	MAE	RMSE	R^2	Decision
Baseline (mean)	$532.26	$622.19	-0.001	Floor — confirms models add real value
Linear Regression	$227.68	$322.32	0.7314	
Random Forest	$151.58	$233.22	0.8594	**Selected**
Gradient Boosting	$163.48	$251.31	0.8367	

Top features driving CLV: income (75.5% of importance), tenure days (8.1%), and total children, recency, age, past campaigns accepted (each 2–3%). Income alone is the dominant signal — most CLV prediction is essentially an income lookup, which is honest but limits how much modelling can add beyond knowing income.

Response classifier — predicting who responds

Model	Recall (Respond)	Recall (Not)	Accuracy	AUC-ROC	Decision
Baseline (random)	0.15	0.85	0.74	0.50	Floor
Logistic Regression	0.7463	0.8218	0.8104	0.8821	**Selected**
Random Forest	0.2687	0.9840	0.8758	0.8722	Highest accuracy — but misses 73% of responders
Gradient Boosting	0.6866	0.8511	0.8262	0.8790	

Why Logistic Regression over Random Forest. Random Forest has the highest accuracy (87.6%), but its recall on the Respond class is only 26.9% — it catches fewer than 1 in 3 actual responders. For a campaign-targeting use case, missing responders is a direct revenue loss. Logistic Regression catches 74.6% of responders at the cost of more false positives — a much better trade-off when contacting a few extra non-responders is cheap.

Targeting priority matrix

CLV tier	Customers	Avg spend	Actual response rate	Predicted probability
High value	737	$1,355	25.1%	44.6%
Mid value	736	$416	12.4%	29.7%
Low value	739	$52	7.7%	23.7%

High-value customers respond at 3.3× the rate of low-value customers (25.1% vs 7.7%) and are worth 26× as much per customer — they are the clear first priority for the next campaign.

Caveats for using the model output

Use the model for ranking, not for absolute probabilities. Predicted probabilities (e.g. 44.6% for high-value tier) are higher than actual response rates (25.1%) because the classifier was trained with balanced sample weights — this maximises recall on the minority class but inflates the predicted probability scale. The order of customers is reliable; the numerical probability is not a calibrated forecast.

Sample bias. The 24 missing-income customers dropped in Phase 2 had a 4.2% response rate vs 15% for the rest. Real-world response for genuinely income-unknown customers may be lower than the model predicts.

What's next

Phase 5 — Sentiment analysis on the Amazon reviews dataset (standalone, not joinable to campaign customers). Phase 6 will use clv_predictions.csv to combine targeting priority with the basket analysis.